



Exercise-1

Marked questions are recommended for Revision.

PART - I : SUBJECTIVE QUESTIONS

Section (A) : VSEPR theory

- A-1. Why NO_2^+ and I_3^- are linear species ?
- A-2. PCl_5 has the shape of a trigonal bipyramidal where as IF_5 has the shape of square pyramidal. Explain.
- A-3. Write the geometry of XeF_4 and OSF_4 using VSEPR theory and clearly indicate the position of lone pair of electrons.
- A-4. Explain the structure of ClF_3 on the basis of VSEPR theory.

Section (B) : Hybridisation

- B-1. Explain hybridisation of central atom in :

(1) XeF_2	(2) XeF_4	(3) PCl_3	(4) PCl_5 (g)
(5) SF_6	(6) IF_3	(7) IF_5	(8) IF_7
(9) CH_4	(10) CCl_4	(11) SiCl_4	(12) SiH_4
(13) H_2O	(14) NH_3	(15) PO_4^{3-}	(16) BrF_5
(17) NO_3^-	(18) CO_3^{2-}	(19) NH_4^+	(20) ClO_3^-

- B-2. The order of size of the hybrid orbitals is as follows $sp < sp^2 < sp^3$. Explain.
- B-3. Draw the structure of the following compounds. Clearly indicate the number of bond pairs and lone pairs involved on central atom. Write (i) number of bond pairs and lone pairs on the central atom (ii) the shape of the molecules (iii) hybridization of the central atom.
(a) SF_4 (b) XeOF_4

Section (C) : Bond angle, bond length comparison

- C-1. Draw an electron dot structure for Br_3^- . Deduce an approximate value of the bond angle.
- C-2. Which compound has the smallest bond angle in each series ?
(a) SbCl_3 SbBr_3 SbI_3
(b) PI_3 AsI_3 SbI_3
- C-3. Compare the C–H bond strength in C_2H_6 , C_2H_4 and C_2H_2 .
- C-4. The POCl_3 molecule has the shape of an irregular tetrahedron with the P atom located centrally. The Cl–P–Cl angle is found to be 103.5° . Give a qualitative explanation for the deviation of this structure from a regular tetrahedron.
- C-5. Which one has highest and least bond angle in the following ?
(1) CH_4 PH_3 AsH_3 SbH_3 (2) H_2O H_2S H_2Te CO_2
(3) PH_3 H_2O (4) Cl_2O ClO_2
(5) PF_3 PH_3 (6) BF_3 NF_3
(7) NH_3 NF_3 (8) PF_3 PCl_3
- C-6. Write the Increasing order of Bond length of each :
(1) C–C, C=C, C≡C (2) C–N, C–O, C–F (3) H–Cl, H–Br, H–I, HF

Section (D) : Multicentered species

- D-1. Find number of $p\pi-d\pi$ bonds in
(a) Disulphate (b) triphosphate (c) trimetaphosphate
(d) trimer of SO_3 (e) P_4O_{10} (f) P_4O_6
- D-2. In which of the following compounds, the $p\pi-d\pi$ bonding take place ?
(a) P_4O_{10} (b) HNO_3 (c) N_2O_5 (d) HClO_3



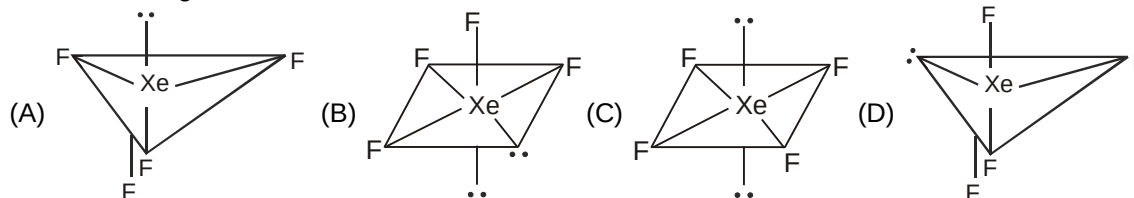
D-3. Calculate individual and average oxidation number (if required) of the marked element and also draw the structure of the following compounds or molecules.

- (1) $\text{Na}_2 \underline{\text{S}}_2 \text{O}_3$ (2) $\text{Na}_2 \underline{\text{S}}_4 \text{O}_6$ (3) $\text{H}_2 \underline{\text{S}} \text{O}_5$ (4) $\text{H}_2 \underline{\text{S}}_2 \text{O}_8$ (5) $\text{H}_2 \underline{\text{S}}_2 \text{O}_7$ (6) $\underline{\text{S}}_8$
 (7) $\text{H} \underline{\text{N}} \text{O}_4$ (8) $\underline{\text{C}}_3 \text{O}_2$ (9) $\underline{\text{O}} \text{S} \text{O}_4$ (10) $\underline{\text{P}} \text{H}_3$ (11) $\underline{\text{Cr}} \text{O}_4^{2-}$ (12) $\underline{\text{Cr}}_2 \text{O}_7^{2-}$
 (13) $\underline{\text{Cr}} \text{O}_2 \text{Cl}_2$ (14) $\underline{\text{Cr}} \text{O}_5$ (15) $\text{Na}_2 \text{H} \underline{\text{P}} \text{O}_4$ (16) $\underline{\text{Fe}} \text{S}_2$ (17) $\underline{\text{C}}_6 \text{H}_{12} \text{O}_6$ (18) $\underline{\text{N}} \text{H}_4 \underline{\text{N}} \text{O}_3$

PART - II : ONLY ONE OPTION CORRECT TYPE

Section (A) : VSEPR theory

A-1. Which is the right structure of XeF_4 ?



A-2. Identify the correct match.

(i) XeF_2	(a) Central atom has sp^3 hybridisation and bent geometry.
(ii) N_3^-	(b) Central atom has $\text{sp}^3 \text{d}^2$ hybridisation and octahedral.
(iii) $\text{PCl}_6^- (\text{PCl}_5 \text{ (s) anion})$	(c) Central atom has sp hybridisation and linear geometry.
(iv) $\text{ICl}_2^+ (\text{I}_2 \text{Cl}_6 \text{ (l) cation})$	(d) Central atom has $\text{sp}^3 \text{d}$ hybridisation and linear geometry.

(A) (i – a), (ii – b), (iii – c), (iv – d) (B) (i – d), (ii – b), (iii – d), (iv – c)
 (C) (i – b), (ii – c), (iii – a), (iv – d) (D) (i – d), (ii – c), (iii – b), (iv – a)

A-3. Which of the following statement is true for $\text{IO}_2 \text{F}_2^-$?

- (A) The electrons are located at the corners of a trigonal bipyramidal but one of the equatorial pairs is unshared.
 (B) It has $\text{sp}^3 \text{d}$ hybridisation and is T-shaped.
 (C) Its structure is analogous to SF_4 .
 (D) (A) and (C) both

A-4. Which reaction involves a change in the electron-pair geometry for the under lined element ?

- (A) $\underline{\text{B}} \text{F}_3 + \text{F}^- \longrightarrow \underline{\text{B}} \text{F}_4^-$ (B) $\underline{\text{N}} \text{H}_3 + \text{H}^+ \longrightarrow \underline{\text{N}} \text{H}_4^+$
 (C) $2 \underline{\text{S}} \text{O}_2 + \text{O}_2 \longrightarrow 2 \underline{\text{S}} \text{O}_3$ (D) $\text{H}_2 \underline{\text{O}} + \text{H}^+ \longrightarrow \text{H}_3 \underline{\text{O}}^+$

A-5. In which of the following molecules number of lone pairs and bond pairs on central atom are not equal ?

- (A) $\text{H}_2 \text{O}$ (B) I_3^- (C) $\text{O}_2 \text{F}_2$ (D) SCl_2

A-6. Which of the following species given below have shape similar to XeOF_4 ?

- (A) XeO_3 (B) IOF_4^+ (C) PCl_5 (D) XeF_5^+

Section (B) : Hybridisation

B-1. The hybridization of carbon atoms in $\text{C}_2\text{--C}_3$ single bond of $\text{HC} \equiv \text{C} - \text{CH} = \text{CH}_2$ is :

- (A) $\text{sp}^3\text{--sp}^3$ (B) $\text{sp}^2\text{--sp}$ (C) sp--sp^2 (D) $\text{sp}^3\text{--sp}$

B-2. Specify the hybridisations of central atom in the following species respectively $\{\text{N}_3^-, \text{NOCl}, \text{N}_2\text{O}\}$

- (A) $\text{sp}, \text{sp}^2, \text{sp}$ (B) $\text{sp}, \text{sp}, \text{sp}^3$ (C) $\text{sp}^2, \text{sp}, \text{sp}$ (D) $\text{sp}^2, \text{sp}^2, \text{sp}$

B-3. In pent-3-en-1-yne the terminal carbon-atoms have following hybridisation

- (A) sp & sp^2 (B) sp^2 & sp^3 (C) sp^2 & sp (D) sp & sp^3

B-4. S_1 : $[\text{XeF}_7]^+$ has $\text{sp}^3 \text{d}^3$ hybridisation

S_2 : $[\text{PCl}_4]^+$ has $\text{sp}^3 \text{d}^2$ hybridisation

S_3 : $[\text{SF}_6]$ has $\text{sp}^3 \text{d}^2$ hybridisation

S_4 : $[\text{PF}_4]^+$ has sp^3 hybridisation

- (A) T F F T (B) T T F T (C) T F T T (D) F F F T



B-5. $\text{BF}_3 + \text{F}^- \rightarrow \text{BF}_4^-$

What is the hybridisation state of B in BF_3 and BF_4^- :

- (A) sp^2 , sp^3 (B) sp^3 , sp^3 (C) sp^2 , sp^2 (D) sp^3 , sp^3d

Section (C) : Bond angle, bond length comparison

C-1. The ONO angle is maximum in :

- (A) HNO_3 (B) NO_2^+ (C) HNO_2 (D) NO_2

C-2. Which statement is correct for N_3^- ion.

- (A) It is bent molecule (B) Bond angle is $< 120^\circ$
(C) Central atom is sp^2 hybridized (D) None of these

C-3. Consider the following molecules ; H_2O H_2S H_2Se H_2Te
I II III IV

Arrange these molecules in increasing order of bond angles.

- (A) $\text{I} < \text{II} < \text{III} < \text{IV}$ (B) $\text{IV} < \text{III} < \text{II} < \text{I}$ (C) $\text{I} < \text{II} < \text{IV} < \text{III}$ (D) $\text{II} < \text{IV} < \text{III} < \text{I}$

C-4. In which of the following bond angle is maximum

- (A) NH_3 (B) NH_4^+ (C) PCl_3 (D) SCl_2

C-5. In which of the following central atom is unhybridised?

- (A) $\text{S}(\text{CH}_3)_2$ (B) SO_2 (C) SiH_4 (D) PCl_3

Section (D) : Multicentered species

D-1. The no. of S-O-S bonds in the trimer of SO_3 is

- (A) 1 (B) 2 (C) 3 (D) None

D-2. Which of the following species do not contain S-S linkage?

- (A) $\text{H}_2\text{S}_2\text{O}_5$ (B) $\text{H}_2\text{S}_2\text{O}_7$ (C) $\text{H}_2\text{S}_2\text{O}_3$ (D) $\text{H}_2\text{S}_4\text{O}_6$

D-3. Which statement is **incorrect** about pyrosilicate ion.

- (A) sp^3 hybridisation
(B) One oxygen atom is shared between two tetrahedron
(C) there are eight Si-O bond
(D) There is one Si-Si bond

D-4. Which is correct about the cyclic silicate $[\text{Si}_6\text{O}_{18}]^{n-}$:

- (A) The value of n is 12
(B) each Si atom is bonded with three oxygen atoms
(C) each oxygen atom is bonded with two Si atoms
(D) all the above are correct.

PART - III : MATCH THE COLUMN

1. Match the following :

	Column-I		Column-II
(A)	SF_2	(p)	sp^3 and bent
(B)	KrF_4	(q)	two lone pairs on central atom
(C)	NOCl	(r)	bond angle $< 109^\circ 28'$
(D)	NF_3	(s)	sp^2 and bent
		(t)	sp^3d^2 and square planar

2. Match the compounds listed in column-I with characteristic(s) listed in column-II.

	Column - I		Column -II
(A)	ClF_2^- , ClF_2^+	(p)	Square pyramidal.
(B)	IO_2F_2^- , F_2SeO	(q)	See-saw and pyramidal shaped respectively.
(C)	IOF_4^- , XeOF_2	(r)	Linear and bent shaped respectively.
(D)	BrF_5 , XeOF_4	(s)	Square pyramidal and T-shaped respectively.
		(t)	Both sp^3d^2 .



3. Match the following :

	Column-I		Column-II
(A)	H ₃ P ₃ O ₉	(p)	S–O–S bond is present
(B)	H ₂ S ₂ O ₇	(q)	Di-basic acid
(C)	H ₂ S ₄ O ₆	(r)	P–O–P bond is present
(D)	H ₄ P ₂ O ₅	(s)	Central atom (S or P) in maximum oxidation state.

Exercise-2

Marked Questions may have for Revision Questions.

PART - I : ONLY ONE OPTION CORRECT TYPE

- The hybridisation of P in phosphate ion (PO₄³⁻) is the same as :
(A) I in ICl₄⁻ (B) S in SO₃ (C) N in NO₃⁻ (D) S in SO₃²⁻
- Choose the molecules in which hybridisation occurs in the ground state ?
(a) BCl₃ (b) NH₃ (c) PCl₃ (d) BeF₂
The correct answer is -
(A) a, b, d (B) a, b, c (C) b, c (D) c, d
- The bent or V-shape of the molecule can be resulted from which of the following hybridization.
(A) sp³ (B) sp² (C) Both (A) and (B) (D) None of these
- sp³d hybridization is considered to be a combination of two hybridization. They are
(A) p³ + sd (B) sp² + pd (C) spd + p² (D) none of these
- If the equatorial plane is x-y plane in sp³d hybridisation then the orbital used in pd hybridisation are -
(A) p_z and d_{z²} (B) p_x and d_{xy} (C) p_y and d_{yz} (D) none of these
- A σ-bonded molecule MX₃ is T-shaped. The number of lone pairs of electrons can be
(A) 0 (B) 2 (C) 1 (D) none of these
- Which of the following should have pyramidal shape :
(A) [ClOF₂]⁺ (B) ICl₃ (C) [BrICl]⁻ (D) All of these
- Which of the following molecules has two lone pairs and bond angle (need not be all bond angles) < 109.5°?
(A) SF₂ (B) KrF₄ (C) ICl₄⁻ (D) All of these
- The correct order of bond angle is :
(A) H₂S < NH₃ < BF₃ < CH₄ (B) NH₃ < H₂S < CH₄ < BF₃
(C) H₂S < NH₃ < CH₄ < BF₃ (D) H₂S < CH₄ < NH₃ < BF₃
- In which of the following molecules are all the bonds not equal?
(A) NF₃ (B) ClF₃ (C) BF₃ (D) AlF₃
- Which of the following is correct order of bond length ?
(A) BF₄⁻ < BF₃ (B) NO₂⁺ < NO₂⁻ (C) CCl₄ < CF₄ (D) ⁺CH₃ > CH₄
- Identify the correct statement :
(A) single N–N bond is stronger than single P–P bond
(B) single N–N bond is weaker than single P–P bond
(C) N≡N is weaker than P≡P
(D) None of these
- In which of the following species peroxide group is not present :
(A) [B₄O₅(OH)₄]²⁻ (B) [S₂O₈]²⁻ (C) CrO₅ (D) HNO₄



14. Which of the following is correct ?
 (A) S_3O_9 : contains no S–S linkage.
 (B) $S_2O_6^{2-}$: contains –O–O– linkage.
 (C) $(HPO_3)_3$: contains P–P linkage
 (D) $S_2O_8^{2-}$: contains S–S linkage
15. The percentage of s-character in the orbital forming P–S bonds in P_4S_3 is :
 (A) 25 (B) 33 (C) 75 (D) 50
16. Indicate the incorrect statement :
 (A) Number of hybrid orbitals formed is equal to no. of atomic orbitals involved.
 (B) $2p_x$ and $2p_y$ - orbitals of carbon can be hybridized to yield two new more stable orbitals
 (C) Effective hybridisation is not possible with orbitals of widely different energies
 (D) The concept of hybridisation has a greater significance in the VB theory of localised orbitals than in the MO theory.
17. In which of the following compounds B atoms are in sp^2 and sp^3 hybridisation states ?
 (A) Borax (B) Diborane (C) Borazole (D) All

PART - II : SINGLE AND DOUBLE VALUE INTEGER TYPE

1. Find the number of planar species
 (a) BF_3 (b) BCl_3 (c) CO_3^{2-} (d) SO_3
 (e) NH_3 (f) NCl_3 (g) PCl_3 (h) XeF_4
2. Find the number of species having bond angle less than $109^\circ 28'$.
 (a) H_2S (b) SO_4^{2-} (c) CCl_4 (d) NH_3
 (e) PH_3 (f) SiH_4 (g) NH_4^+ (h) PF_3
 (i) NH_2^- (j) SO_3 (k) H_2O
3. Find out total number of π bond in following xenon compounds.
 $XeOF_2$, XeO_2F_4 , XeO_3 , XeO_4 , XeO_3F_2 , $XeOF_4$, XeO_2F_2
4. P_4O_{10} has two different types of P–O bonds. Find the no. of P–O bonds with shorter bond length.
5. Difference in the oxidation number of sulphur atom is in $Na_2S_4O_6$ is x, that of $H_2S_2O_5$ is y. Find value of $x \times y$ is :
6. In a P_4O_6 molecule, the total number of P–O–P bonds is :

PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

1. Which statement is correct about hybridization ?
 (A) In hybridisation orbitals take part.
 (B) In hybridisation electrons take part.
 (C) In hybridisation fully filled, half filled or empty orbitals can take part.
 (D) Hybridised orbitals only contains bond pair electron.
2. Which of the following represent the given mode of hybridisation $sp^2-sp^2-sp-sp$ from left to right
 (A) $H_2C=C=CH_2$ (B) $HC\equiv C-C\equiv CH$
 (C) $H_2C=CH-C\equiv N$ (D) $H_2C=CH-C\equiv CH$
3. Which is/are in linear shape ?
 (A) NO_2^+ (B) XeF_2 (C) I_3^- (D) I_3^+
4. Which is true about NH_2^- , NH_3 , NH_4^+ ?
 (A) Hybridization of N is same. (B) No. of lone pair of electron on N are same.
 (C) Molecular geometry (i.e. shape) is different. (D) Bond angle is same.
5. Which of the following molecule (s) has/have bond angle close to 90° ?
 (A) NH_3 (B) H_2S (C) PH_3 (D) ICl_3



6. Which of the following statements are true about borax :
- Boron atoms are present in 2 different oxidation state and it different by 1
 - the average oxidation state of boron is same that in B_2H_6 .
 - Boron atoms are present in different hybridization
 - 2 boron atoms are connected with 4 oxygen each and 2 boron atoms are connected with 3 oxygen atom each.
7. Identify the correct statement
- $H_2S_2O_7$ has peroxy linkage
 - $H_2S_2O_6$ has S-S linkage
 - $H_2S_2O_8$ has peroxy linkage
 - H_2SO_3 (Sulphurous acid) has S in +4 oxidation state
8. In which of the following compound(s) oxidation number of one central atom is/are ≥ 6 ?
- N_2O_6
 - CrO_5
 - H_3PO_5
 - $H_2S_2O_8$

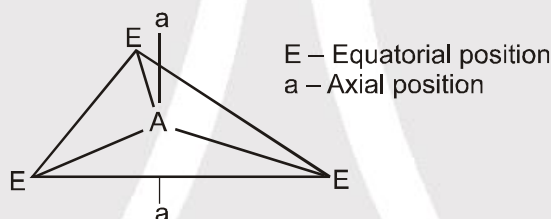
PART - IV : COMPREHENSION

Read the following passage carefully and answer the questions.

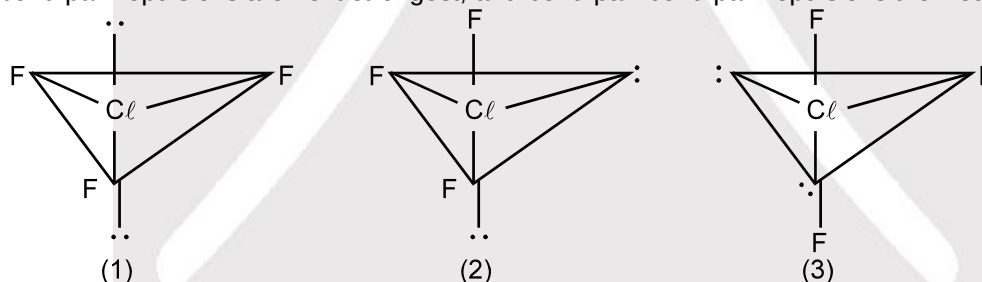
Comprehension # 1

VSEPR THEORY

The trigonal bipyramid is not a regular shape since the bond angles are not all the same. It therefore follows that the corners are not equivalent in ClF_3 molecule. Lone pairs occupy two of the corners, and F atoms occupy the other three corners. These different arrangements are theoretically possible, as shown in figure.



(i) The most stable structure will be the one of lowest energy, that is the one with the minimum repulsion between the five orbitals. The greatest repulsion occurs between two lone pairs. Lone pair bond pair repulsions are next strongest, and bond pair-bond pair repulsions the weakest.



A rule of thumb can be theorised, that the position having maximum repulsion amongst them are occupied at equatorial points. Therefore (3) structure is right.

(ii) Since double bond occupies more space compared to single bond therefore it will prefer equatorial position.

(iii) More electronegative element will occupy axial position in case of trigonal bipyramidal geometry

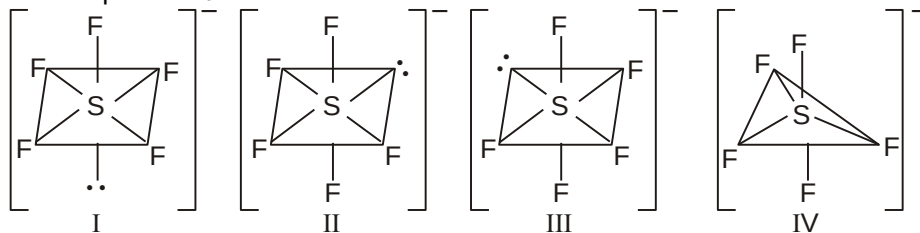
(iv) In case of sp^3d^2 hybridisation lone pairs should be placed opposite to each other because all the corners are identical.

1. Geometry (i.e. arrangement of electron pairs around central atom) of ClO_3F is similar to the :
- XeF_4
 - $SOCl_2$
 - I_3^-
 - ClO_4^-





2. The shape of SF_5^- can be :



(A) I only

(B) I and II only

(C) IV only

(D) I, II, & III

3. Actual shape of the molecule BrF_5 is similar to the molecule :

(A) PCl_5

(B) XeF_4

(C) PCl_4^+

(D) None of these

4.* Which of the following do not exist ?

(A) SH_6

(B) HFO_4

(C) Sl_6

(D) HClO_3

Comprehension # 2

Answer Q.5, Q.6 and Q.7 by appropriately matching the information given in the three columns of the following table.

Column-1 (Compound)		Column-2 (Shape)		Column-3 (Hybridisation)	
(I)	XeF_4	(i)	Tetrahedral	(P)	sp^3
(II)	ClF_3	(ii)	Square planar	(Q)	sp^2
(III)	SiF_4	(iii)	Bent	(R)	sp^3d
(IV)	CH_3OCH_3	(iv)	T-shape	(S)	sp^3d^2

5. Which of the following combination is true for compound which have 2 lone pair of electrons on central atom?

(A) (I), (ii), (S)

(B) (I), (ii), (R)

(C) (I), (i), (P)

(D) (II), (iv), (Q)

6. Which combination is correct for the compound having bond angle $> 109^\circ 28'$?

(A) (III), (i), (P)

(B) (IV), (iii), (P)

(C) (IV), (ii), (P)

(D) (I), (ii), (S)

7. Which of the following is true for a planar compound ?

(A) (III), (i), (P)

(B) (I), (iv), (P)

(C) (II), (iv), (R)

(D) (II), (i), (P)

Exercise-3

PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

* Marked Questions may have more than one correct options.

- The hybridization of atomic orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ are : [JEE-2000(S), 1/135]
 (A) sp , sp^3 and sp^2 respectively (B) sp , sp^2 and sp^3 respectively
 (C) sp^2 , sp and sp^3 respectively (D) sp^2 , sp^3 and sp respectively
- The number of P—O—P bonds in tricyclic metaphosphoric acid is : [JEE-2000(S), 1/135]
 (A) zero (B) two (C) three (D) four
- Draw the molecular structures of XeF_2 , XeF_4 and XeO_2F_2 , indicating the location of lone pair of electrons. [JEE-2000(M), 3/135]



4. The correct order of hybridisation of the central atom in the following species; NH_3 , PCl_5 and BCl_3 is : [JEE-2001(S), 1/135]
 (A) dsp^2 , sp^2 , sp^3 (B) sp^3 , dsp^3 , sp^2 (C) dsp^2 , sp^3 , dsp^3 (D) dsp^2 , sp^2 , dsp^3
5. The number of S—S bonds, in sulphur trioxide trimer (S_3O_9) is : [JEE-2001(S), 1/135]
 (A) three (B) two (C) one (D) Zero
6. Which of the following are isoelectronic and isostructural ? [JEE-2003(S), 3/144]
 NO_3^- , CO_3^{2-} , ClO_3^- , SO_3
 (A) NO_3^- , CO_3^{2-} (B) SO_3 , NO_3^- (C) ClO_3^- , CO_3^{2-} (D) CO_3^{2-} , SO_3
7. Using VSEPR theory, draw the shape of PCl_5 and BrF_5 . [JEE-2003(M), 2/144]
8. Use VSEPR model to draw the structures of OSF_4 and XeF_4 (indicate the lone pair(s) on central atom) and specify their geometry. [JEE-2004(M), 2/144]
9. The percentage of p-character in the orbitals forming P—P bonds in P_4 is : [JEE-2007, 3/162]
 (A) 25 (B) 33 (C) 50 (D) 75
- 10.* The nitrogen oxide(s) that contain(s) N—N bond(s) is(are) : [JEE-2009, 4/160]
 (A) N_2O (B) N_2O_3 (C) N_2O_4 (D) N_2O_5
11. The species having pyramidal shape is : [IIT-JEE-2010, 5/163]
 (A) SO_3 (B) BrF_3 (C) SiO_3^{2-} (D) OSF_2
12. Based on VSEPR theory, the number of 90 degree F—Br—F angles in BrF_5 is : [JEE-2010, 3/163]
13. The shape of XeO_2F_2 molecule is [JEE-2012, 3/136]
 (A) trigonal bipyramidal (B) square planar
 (C) tetrahedral (D) see-saw
14. The total number of lone pairs of electrons in N_2O_3 is : [JEE(Advanced) 2015, 4/168]
15. Among the triatomic molecules/ions, BeCl_2 , N_3^- , N_2O , NO_2^+ , O_3 , SCl_2 , ICl_2^- , I_3^- and XeF_2 , the total number of linear molecules(s)/ion(s) where the hybridization of the central atom does not have contribution from the d-orbital(s) is [JEE(Advanced) 2015, 4/168]
 [Atomic number : S = 16, Cl = 17, I = 53 and Xe = 54]
- 16.* The crystalline form of borax has [JEE Advanced 2016, 4/124]
 (A) tetranuclear $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$ unit
 (B) all boron atoms in the same plane
 (C) equal number of sp^2 and sp^3 hybridized boron atoms
 (D) one terminal hydroxide per boron atom
17. The total number of compounds having at least one bridging oxo group among the molecules given below is _____. [JEE Advanced 2018, 3/120]
 N_2O_3 , N_2O_5 , P_4O_6 , P_4O_7 , $\text{H}_4\text{P}_2\text{O}_5$, $\text{H}_5\text{P}_3\text{O}_{10}$, $\text{H}_2\text{S}_2\text{O}_3$, $\text{H}_2\text{S}_2\text{O}_5$



PART - II : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

JEE(MAIN) OFFLINE PROBLEMS

1. The hybridisation of the underline atom changes in : [AIEEE-2002, 3/225]
 (1) AlH₃ changes to AlH₄⁻ (2) H₂O changes to H₃O⁺
 (3) NH₃ changes to NH₄⁺ (4) in all cases
2. Bond angle of 109° 28' is found in : [AIEEE-2002, 3/225]
 (1) NH₃ (2) H₂O (3) CH₃⁺ (4) NH₄⁺
3. Which of the following compounds has the smallest bond angle in its molecule ? [AIEEE-2003, 3/225]
 (1) SO₂ (2) H₂O (3) H₂S (4) NH₃
4. The pair of species having identical shapes for molecules of both species is : [AIEEE-2003, 3/225]
 (1) CF₄, SF₄ (2) XeF₂, CO₂ (3) BF₃, PCl₃ (4) PF₅, IF₅.
5. The maximum number of 90° angles between bond pair–bond pair of electrons is observed in : [AIEEE-2004, 3/225]
 (1) dsp³ (2) sp³d (3) dsp² (4) sp³d²
6. The correct order of bond angles (smallest first) in H₂S, NH₃, BF₃ and SiH₄ is : [AIEEE-2004, 3/225]
 (1) H₂S < SiH₄ < NH₃ < BF₃ (2) NH₃ < H₂S < SiH₄ < BF₃
 (3) H₂S < NH₃ < SiH₄ < BF₃ (4) H₂S < NH₃ < BF₃ < SiH₄
7. The molecular shapes of SF₄, CF₄ and XeF₄ are : [AIEEE-2005, 3/225]
 (1) the same with 2, 0 and 1 lone pairs of electrons on the central atom, respectively.
 (2) the same with 1, 1 and 1 lone pair of electrons on the central atom, respectively.
 (3) different with 0, 1 and 2 lone pairs of electrons on the central atom, respectively.
 (4) different with 1, 0 and 2 lone pairs of electrons on the central atom, respectively.
8. The hybridisation of orbitals of N atom in NO₃⁻, NO₂⁺ and NH₄⁺ are respectively : [AIEEE-2011, 4/120]
 (1) sp, sp², sp³ (2) sp², sp, sp³ (3) sp, sp³, sp² (4) sp², sp³, sp
9. The structure of IF₇ is : [AIEEE-2011, 4/120]
 (1) square pyramid (2) trigonal bipyramid (3) octahedral (4) pentagonal bipyramid
10. The molecule having smallest bond angle is : [AIEEE-2012, 4/120]
 (1) NCl₃ (2) AsCl₃ (3) SbCl₃ (4) PCl₃
11. In which of the following pairs the two species are not isostructural ? [AIEEE-2012, 4/120]
 (1) CO₃²⁻ and NO₃⁻ (2) PCl₄⁺ and SiCl₄ (3) PF₅ and BrF₅ (4) AlF₆³⁻ and SF₆
12. The species in which the N atom is in a state of sp hybridization is : [JEE(Main)-2016, 4/120]
 (1) NO₂⁻ (2) NO₃⁻ (3) NO₂ (4) NO₂⁺

JEE(MAIN) ONLINE PROBLEMS

1. Which one of the following does not have a pyramidal shape? [JEE(Main) 2014 Online (11-04-14), 4/120]
 (1) (CH₃)₃N (2) (SiH₃)₃N (3) P(CH₃)₃ (4) P(SiH₃)₃



2. The geometry of XeOF_4 by VSEPR theory is : [JEE(Main) 2015 Online (10-04-15), 4/120]
 (1) pentagonal planar (2) octahedral (3) square pyramidal (4) trigonal bipyramidal
3. Which of the following compound has a P–P bond ? [JEE(Main) 2015 Online (11-04-15), 4/120]
 (1) $\text{H}_4\text{P}_2\text{O}_5$ (2) $(\text{HPO}_3)_3$ (3) $\text{H}_4\text{P}_2\text{O}_6$ (4) $\text{H}_4\text{P}_2\text{O}_7$
4. Choose the incorrect formula out of the four compounds for an element X below :
 (1) X_2O_3 (2) X_2Cl_3 (3) $\text{X}_2(\text{SO}_4)_3$ (4) XPO_4
5. The group of molecules having identical shape is : [JEE(Main) 2016 Online (09-04-16), 4/120]
 (1) PCl_5 , IF_5 , XeO_2F_2 (2) BF_3 , PCl_3 , XeO_3
 (3) ClF_3 , XeOF_2 , XeF_3^+ (4) SF_4 , XeF_4 , CCl_4
6. **Assertion:** Among the carbon allotropes, diamond is an insulator, Whereas, graphite is a good conductor of electricity.
Reason: Hybridization of carbon in diamond and graphite are sp^3 and sp^2 , respectively. [JEE(Main) 2016 Online (10-04-16), 4/120]
 (1) Both assertion and reason are correct, but the reason is not the correct explanation for the assertion.
 (2) Assertion is incorrect statement, but the reason is correct.
 (3) Both assertion and reason are correct, and the reason is the correct explanation for the assertion.
 (4) Both assertion and reason are incorrect.
7. The bond angle H–X–H is the greatest in the compound : [JEE(Main) 2016 Online (10-04-16), 4/120]
 (1) NH_3 (2) PH_3 (3) CH_4 (4) H_2O
8. Identify the pair in which the geometry of the species is T-shape and square-pyramidal, respectively : [JEE(Main) 2018 Online (15-04-18), 4/120]
 (1) ICl_2^- and ICl_5 (2) IO_3^- and IO_2F_2^-
 (3) ClF_3 and IO_4^- (4) XeOF_2 and XeOF_4
9. In graphite and diamond, the percentage of p-characters of the hybrid orbitals in hybridization are respectively : [JEE(Main) 2018 Online (15-04-18), 4/120]
 (1) 33 and 25 (2) 67 and 75 (3) 50 and 75 (4) 33 and 75
10. The decreasing order of bond angles in BF_3 , NH_3 , PF_3 and I_3^- is :
 (1) $\text{I}_3^- > \text{BF}_3 > \text{NH}_3 > \text{PF}_3$ (2) $\text{BF}_3 > \text{I}_3^- > \text{PF}_3 > \text{NH}_3$
 (3) $\text{BF}_3 > \text{NH}_3 > \text{PF}_3 > \text{I}_3^-$ (4) $\text{I}_3^- > \text{NH}_3 > \text{PF}_3 > \text{BF}_3$
11. The number of P–O bonds in P_4O_6 is : [JEE(Main) 2018 Online (15-04-18), 4/120]
 (1) 6 (2) 9 (3) 12 (4) 18
12. In XeO_3F_2 , the number of bond pair(s), π -bond(s) and lone pair(s) on Xe atom respectively are : [JEE(Main) 2018 Online (15-04-18), 4/120]
 (1) 5, 2, 0 (2) 4, 2, 2 (3) 5, 3, 0 (4) 4, 4, 0
13. Among the oxides of nitrogen : [JEE(Main) 2018 Online (16-04-18), 4/120]
 N_2O_3 , N_2O_4 and N_2O_5 ; the molecule(s) having nitrogen-nitrogen bond is/are :
 (1) N_2O_3 and N_2O_4 (2) N_2O_4 and N_2O_5 (3) N_2O_3 and N_2O_5 (4) Only N_2O_5



14. Which of the following conversions involves change in both shape and hybridisation ?
[JEE(Main) 2018 Online (16-04-18), 4/120]
(1) $\text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+$ (2) $\text{BF}_3 \rightarrow \text{BF}_4^-$ (3) $\text{CH}_4 \rightarrow \text{C}_2\text{H}_6$ (4) $\text{NH}_3 \rightarrow \text{NH}_4^+$
15. The **incorrect** geometry is represented by :
[JEE(Main) 2018 Online (16-04-18), 4/120]
(1) NF_3 – trigonal planar (2) BF_3 – trigonal planar
(3) AsF_5 – trigonal bipyramidal (4) H_2O – bent
16. The type of hybridisation and number of lone pair(s) of electrons of Xe in XeOF_4 , respectively, are :
[JEE(Main) 2019 Online (10-01-19), 4/120]
(1) sp^3d^2 and 1 (2) sp^3d^2 and 2 (3) sp^3d and 1 (4) sp^3d and 2
17. The pair that contains two P–H bonds in each of the oxoacid is:
[JEE(Main) 2019 Online (10-01-19), 4/120]
(1) $\text{H}_4\text{P}_2\text{O}_5$ and H_3PO_3 (2) $\text{H}_4\text{P}_2\text{O}_5$ and $\text{H}_4\text{P}_2\text{O}_6$
(3) H_3PO_2 and $\text{H}_4\text{P}_2\text{O}_5$ (4) H_3PO_3 and H_3PO_2
18. The element that shows greater ability to form $\text{p}\pi\text{-p}\pi$ multiple bonds, is:
[JEE(Main) 2019 Online (12-01-19), 4/120]
(1) Ge (2) Sn (3) C (4) Si





Answers

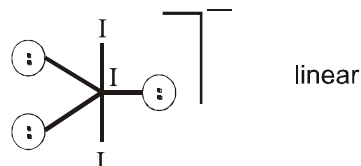
EXERCISE - 1

PART - I

A-1.

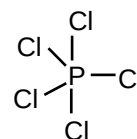
In NO_2^+ the N has sp hybridisation; so it is linear $\text{O}=\text{N}^+=\text{O}$

In I_3^- there are 5 electron pairs around central iodine atom (3 lone pairs and 2 bond pairs). The hybridisation of iodine is thus sp^3d . To have minimum repulsions between lp-lp and lp-bp it acquires linear shape as shown below.

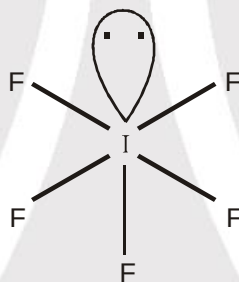


A-2.

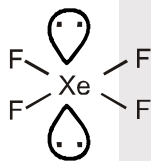
In PCl_5 there are 5 electron pairs around central phosphorus atom and all are bond pairs. The hybridisation of phosphorus is thus sp^3d . To have minimum repulsions between bp-bp it acquires trigonal bipyramidal shape as shown below.



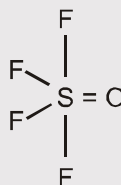
In IF_5 there are 6 electron pairs around central iodine atom. The hybridisation of iodine is thus sp^3d^2 . 6 electron pairs contain 5 bond pairs and one lone pair so it will be square pyramidal to have minimum repulsions between lp-bp and bp-bp.



A-3.

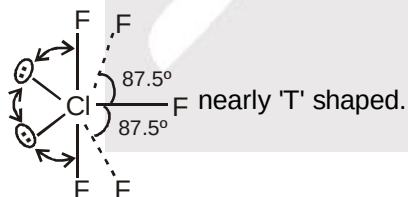


Square planar



Trigonal bipyramidal

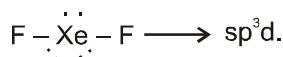
A-4.



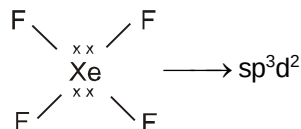
F nearly 'T' shaped.

B-1.

1.



2.



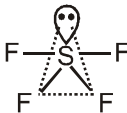
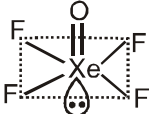


3. $\begin{array}{c} \text{Cl} \\ | \\ \text{P} \\ | \\ \text{Cl} \end{array} \text{Cl} \longrightarrow sp^3$
4. $\begin{array}{c} \text{Cl} \\ | \\ \text{Cl}-\text{P}-\text{Cl} \\ | \\ \text{Cl} \end{array} \longrightarrow sp^3d$
5. $\begin{array}{c} \text{F} \\ | \\ \text{F}-\text{S}-\text{F} \\ | \\ \text{F} \end{array} \longrightarrow sp^3d^2$
6. $\text{IF}_3 : - \text{F} \longrightarrow sp^3d$
7. $\text{IF}_5 : - \text{F} \longrightarrow sp^3d^2$
8. $\text{IF}_7 : - \text{F} \longrightarrow sp^3d^3$
9. $\text{CH}_4 : - \longrightarrow sp^3$
10. $\text{CCl}_4 : - \longrightarrow sp^3$
11. $\text{SiCl}_4 : - \longrightarrow sp^3$
12. $\text{SiH}_4 : - \longrightarrow sp^3$
13. $\text{H}_2\text{O} : - \text{H} \longrightarrow sp^3$
14. $\text{NH}_3 : - \text{H} \longrightarrow sp^3$
15. $\begin{array}{c} \text{O} \\ || \\ \text{P} \\ | \\ \text{O} \end{array} \text{O} \longrightarrow sp^3$
16. $\text{BrF}_5 : - \longrightarrow sp^3d^2$
17. $\text{O}=\text{N}-\text{O}^- \longrightarrow sp^2$
18. $\text{O}=\text{C}-\text{O}^- \longrightarrow sp^2$
19. $\text{H}-\text{N}^+-\text{H} \longrightarrow sp^3$
20. $\begin{array}{c} \text{O} \\ || \\ \text{Cl}-\text{O} \\ || \\ \text{O} \end{array} \longrightarrow sp^3$

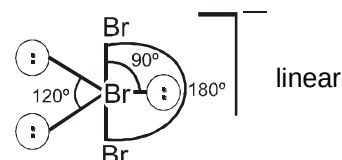
B-2. True, As the s character in hybrid orbital decrease, size of hybrid orbital increases.



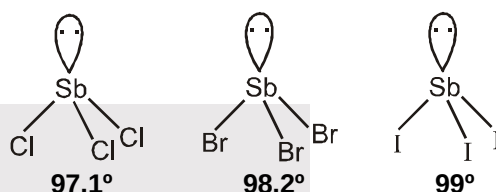


- B-3.** (a) SF_4  4 bond pair & 1 lone pair, Hybridization = sp^3d Shape : see saw
- (b) XeOF_4  5 bond pair & 1 lone pair, Hybridization = sp^3d^2 Shape : Square pyramidal

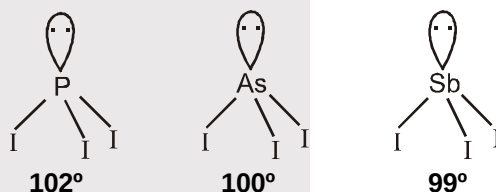
- C-1.** In Br_3^- there are 5 electron pairs around central bromine atom (3 lone pairs and 2 bond pairs). The hybridisation of bromine is thus sp^3d . To have minimum repulsions between lp-lp and lp-bp it acquires linear shape as shown below.



- C-2.** (a) Cl, the most electronegative of the halogens in this series, pulls shared electrons the most strongly away from Sb, reducing electron density near Sb. The consequence is that the lone pair exerts the strongest influence on shape in SbCl_3 .

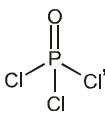


- (b) Phosphorus is the most electronegative of the central atoms. Consequently, it exerts the strongest pull on shared electrons, concentrating these electrons near P and increasing bonding pair-bonding pair repulsions—hence, the largest angle in PI_3 . Sb, the least electronegative central atoms, has the opposite effect: Shared electrons are attracted away from Sb, reducing repulsions between the Sb-I bonds.



The consequence is that the effect of the lone pair is greatest in SbI_3 , which has the smallest angle. Atomic size arguments can also be used for these species. Larger outer atoms result in larger angles; larger central atoms result in smallest angles.

- C-3.** Order of C-H bond strength is $\text{C}_2\text{H}_2 > \text{C}_2\text{H}_4 > \text{C}_2\text{H}_6$ as %s character decreases in the same order.

- C-4.**  double bond occupies large area and has large electron density. So there is intrinsic repulsion between P=O and P-Cl bond pairs. To minimize this repulsion bond angle decrease from 109.5° to 103.5° .

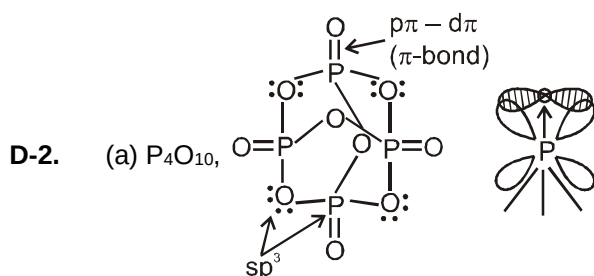
- | C-5. Highest bond angle | Lowest bond angle |
|--|--|
| (1) CH_4 | SbH_3 |
| (2) CO_2 | H_2Te |
| (3) H_2O | PH_3 |
| (4) ClO_2 | Cl_2O |
| (5) PF_3 (sp^3 hybridisation) | PH_3 (no hybridisation) |
| (6) BF_3 (sp^2 hybridisation) | NF_3 (sp^3 hybridisation) |
| (7) NH_3 | NF_3 |
| (8) PCl_3 | PF_3 |

- C-6.** (1) $\text{C}\equiv\text{C} < \text{C}=\text{C} < \text{C}-\text{C}$ (2) $\text{C}-\text{F} < \text{C}-\text{O} < \text{C}-\text{N}$ (3) $\text{HF} < \text{H}-\text{Cl} < \text{H}-\text{Br} < \text{H}-\text{I}$

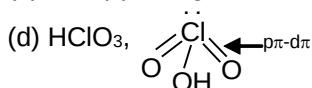




D-1. (a) 4 (b) 3 (c) 3 (d) 6 (e) 4 (f) 0



(b) and (c) nitrogen does not have empty d-orbital.



D-3. (1) +2 (6, -2) (2) +5/2 (5, 5, 0, 0) (3) +6 (4) +6 (+6, +6) (5) +6 (+6, +6)
 (6) 0 (7) +5 (8) 4/3 (+2, +2, 0) (9) +8 (10) -3
 (11) +6 (12) +6 (+6, +6) (13) +6 (14) +6 (15) +5
 (16) +2 (17) 0 (18) -3, +5

Note: Inside the bracket () answer for individual oxidation number is given.

PART - II

A-1. (C)	A-2. (D)	A-3. (D)	A-4. (A)	A-5. (B)
A-6. (D)	B-1. (B)	B-2. (A)	B-3. (D)	B-4. (C)
B-5. (A)	C-1. (B)	C-2. (D)	C-3. (B)	C-4. (B)
C-5. (A)	D-1. (C)	D-2. (B)	D-3. (D)	D-4. (A)

PART - III

1. (A - p, q, r) ; (B - q, r, t) ; (C - s) ; (D - r) 2. (A - r) ; (B - q) ; (C - s) ; (D - p, t)
 3. (A - r, s) ; (B - p, q, s) ; (C - q) ; (D - q, r)

EXERCISE - 2

PART - I

1. (D)	2. (C)	3. (C)	4. (B)	5. (A)
6. (B)	7. (A)	8. (D)	9. (C)	10. (B)
11. (B)	12. (B)	13. (A)	14. (A)	15. (A)
16. (B)	17. (A)			

PART - II

1. 5 (a, b, c, d, h)	2. 6 (a, d, e, h, i, k)	3. 16
4. 4	5. 10	6. 6



PART - III

- | | | | | |
|----------|----------|----------|---------|----------|
| 1. (AC) | 2. (CD) | 3. (ABC) | 4. (AC) | 5. (BCD) |
| 6. (BCD) | 7. (BCD) | 8. (BD) | | |

PART - IV

- | | | | | |
|--------|--------|--------|-----------|--------|
| 1. (C) | 2. (D) | 3. (D) | 4.* (ABC) | 5. (A) |
| 6. (B) | 7. (C) | | | |

EXERCISE - 3

PART - I

1. (B) 2. (C)

3. According to VSEPR theory

Number of electron pairs = 5,

Number of bond pairs = 2,

So, Number of lone pairs = 3.

Thus XeF_2 is linear with 3 lone pairs occupying 3 equatorial positions of trigonal bipyramidal so as to minimize the repulsions.

Number of electron pairs = 6,

Number of bond pairs = 4,

So, Number of lone pairs = 2.

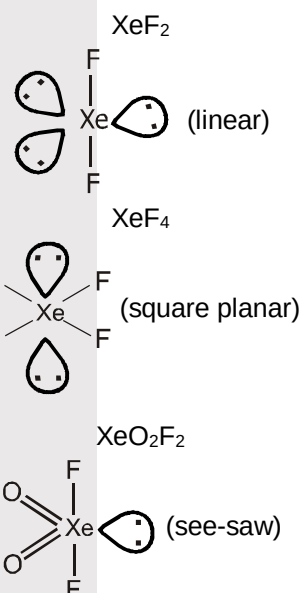
Thus XeF_4 is linear with 2 lone pairs occupying 2 axial positions of octahedral pyramidal so as to minimize the repulsions.

Number of electron pairs (including super electron pairs) = 5,

Number of bond pairs = 4,

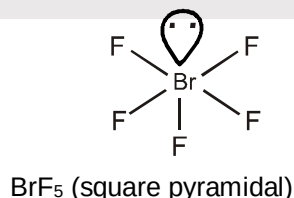
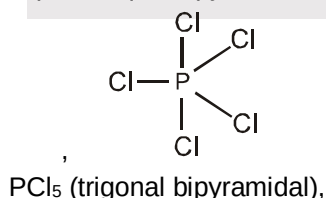
So, Number of lone pairs = 1.

Thus XeO_2F_2 is see-saw with 1 lone pairs occupying one equatorial position and two double bonds occupying other two equatorial positions of trigonal bipyramidal so as to minimize the repulsions.



4. (B) 5. (D) 6. (A)

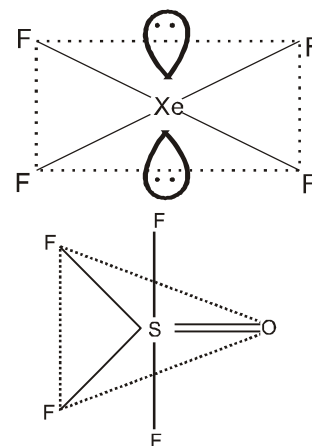
7. There are 5 electron pairs and all are bonds pairs in PCl_5 . So to have the minimum repulsions between bond pairs it acquires trigonal bipyramidal shape. In BrF_5 , there are 6 electrons pairs out of which one lone pair and rest all are bond pairs. So to have the minimum repulsions between bond pairs and lone pairs it acquires square pyramidal shape.





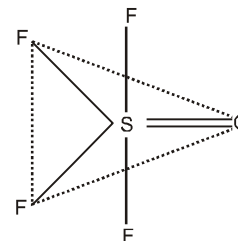
8.

According to VSEPR theory two lone pairs out of six electron pairs are trans to each other to have minimum repulsion. The shape of XeF_4 is square planar and geometry is octahedral with sp^3d^2 hybridisation. The molecule looks like :



In OSF_4 , there are five electron pairs and all are bond pairs. So geometry is trigonal bipyramidal. As double bond creates more repulsion than single bond, the double bond acquires one of equatorial position of trigonal bipyramidal to have minimum repulsions.

The structure looks like:



9. (D) 10.* (ABC) 11. (D) 12. 0 or 8 13. (D)
 14. 8 14. 4 16.* (ACD) 17. 5 or 6

PART - II

JEE(MAIN) OFFLINE PROBLEMS

- | | | | | |
|---------|---------|--------|--------|---------|
| 1. (1) | 2. (4) | 3. (3) | 4. (2) | 5. (4) |
| 6. (3) | 7. (4) | 8. (2) | 9. (4) | 10. (3) |
| 11. (3) | 12. (4) | | | |

JEE(MAIN) ONLINE PROBLEMS

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (2) | 2. (3) | 3. (3) | 4. (2) | 5. (3) |
| 6. (1) | 7. (3) | 8. (4) | 9. (2) | 10. (1) |
| 11. (3) | 12. (3) | 13. (1) | 14. (2) | 15. (1) |
| 16. (1) | 17. (3) | 18. (3) | | |

